

Ganesh Neelakanta

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Professional Summary

AI Engineer with 3.5 years designing and deploying production-grade agentic AI, generative AI, and retrieval solutions – LLM agents with tool/function calling, multi-step orchestration, and guardrails; **Databricks** (Vector Search, Unity Catalog, MLflow); **RAG/Document AI** (OCR, chunking, embeddings, hybrid search, reranking); and **API** integrations across Azure OpenAI, Gemini, and Claude. Strong stakeholder engagement translating business needs into requirements, with 85% reduction in manual effort delivered and ownership of evaluation, monitoring, and incident resolution in production.

Experience

Deloitte

Generative AI Consultant

Hyderabad, Telangana

November 2025 – Present

- Designed **SOWnia**, an AI-powered Statement of Work review system featuring 8 specialized AI review agents (Financial, Legal/Risk, Technical, Delivery, Organizational, Strategic domains) using **Claude Opus**, **LangChain**, and **MCP tool/function-calling integrations**; implemented confidence-based routing, contradiction detection, and risk flagging with page-level traceability and CSV/PDF audit-trail exports – replacing multi-expert, multi-day review cycles with structured findings in minutes
- Directly engaged business and technical stakeholders to gather requirements, define acceptance criteria, and translate findings into practical recommendations, authoring technical designs, test plans, and status updates with minimal oversight
- Designed **context engineering strategies** (dynamic context assembly, compression, conversation memory) to maintain continuity across long agent sessions
- Implemented **guardrails architecture** (input validation, output filtering, PII redaction, audit logging) and an **agent evaluation framework** (accuracy, grounding, task completion) to measure production reliability

Anblicks

AI and Data Engineer

Hyderabad, Telangana

June 2023 – October 2025

- Developed **4-stage multi-agent LLM pipeline** for automated marketing copy generation with data extraction, Gemini-powered attribute enrichment, Azure OpenAI content generation, and **LLM-as-Judge** compliance refinement; produced tailored content for 3 audience segments across Email, SMS, and Mobile channels, achieving **85% reduction in manual effort** and cutting content creation time from hours to minutes
- Implemented **LLM-as-Judge** pattern for automated brand compliance enforcement with Snowflake caching layer to prevent repeated content across campaign runs; applied **prompt versioning and structured output formatting** to systematically improve agent response quality
- Established 3 production **Agentic RAG** knowledge assistants for HR, Supply Chain, and Legal by ingesting PPTX/PDF via Azure Document Intelligence (**OCR, layout/table extraction**) with 3-tier extraction and PyMuPDF; applied slide-level, semantic, and fixed-size chunking; used **Databricks Vector Search** (hybrid search, query rewriting, reranking, metadata filtering), **RAGAS evaluation**, AES-256 encrypted chat history, and FastAPI/AD-authenticated frontends
- Configured **BGE-Large-EN** and **text-embedding-3-large** embeddings, tuning model and chunking choices for relevance, latency, and cost; built **LangChain LCEL RAG chains**; versioned models via **MLflow** with **Unity Catalog** registration and **LangSmith** tracing
- Deployed and monitored services on **Azure and Databricks** with defined container CI/CD, and ownership of data/model/orchestration troubleshooting
- Applied **time series forecasting** (ARIMA, SARIMA, XGBoost) and **anomaly detection** for predictive analytics on Snowflake credit usage, reducing cloud costs

Data Engineering Trainee

March 2023 – May 2023

- Completed intensive training in Python, SQL, Snowflake, Azure Data Factory, Power BI, and data warehouse design with focus on scalable ETL pipeline design
- Built end-to-end ETL pipeline consolidating multi-source retail sales data via Azure Data Factory into Snowflake; delivered 4 automated Power BI dashboards across product category, customer segment, regional, and time-series dimensions, leading to a full-time offer within 3 months

Key Projects

SOWnia – AI-Powered SOW Multi-Agent Review System

November 2025 – Present

Claude Opus, LangChain, MCP, FastAPI, React, Azure Document Intelligence, LangSmith

- Deployed 8-agent SOW review system with **human-in-the-loop (HITL)** checkpoints, **tool/function calling**, and **confidence-based routing**; users upload PDF/DOCX, select a review angle, and receive page-referenced findings with contradiction detection and risk flags; implemented **agent evaluation framework** with grading rubrics to measure accuracy and task completion rates; exports to CSV/PDF reducing review turnaround from days to minutes

AI-Powered Creative Copy Multi-Agent System

November 2024 – April 2025

Azure OpenAI, Google Gemini, LangChain, LangGraph, Snowflake, FastAPI, React

- Implemented 4-stage multi-agent pipeline with **LangGraph**-orchestrated decision branching and error recovery; Gemini-powered attribute enrichment, Azure OpenAI content generation, and LLM-as-Judge compliance refinement; produced brand-compliant copy across Email, SMS, and Mobile channels with Snowflake anti-repetition caching

AI Voice-Over Multilingual TTS Pipeline

March 2025 – June 2025

Google Cloud Translate API v2, Chirp3-HD, Azure Speech, Databricks, FastAPI

- Designed modular TTS architecture with thread-based concurrent synthesis, provider-switching between Azure Speech and Databricks modules, and a phonetic placeholder engine protecting brand-critical terms from translation distortion across 130+ languages; deployed as **FastAPI microservice** with REST API integration

Advanced RAG Knowledge Assistants for Enterprise

July 2024 – October 2024

Databricks, LangChain LCEL, BGE-Large-EN, text-embedding-3-large, Azure Document Intelligence, MLflow, LangSmith, FastAPI, React

- Deployed three production agentic RAG chatbots for HR, Supply Chain, and Legal domains using 3-tier document extraction and OCR with PyMuPDF, slide-level and semantic chunking, **Databricks Vector Search** with hybrid search and query rewriting, **RAGAS evaluation scoring**, AES-256 encrypted chat history, Active Directory authentication, and MLflow-versioned scale-to-zero model serving with React and FastAPI frontends

GenAI Sentiment Analysis – Voice of Customer Dashboard

DSPy, PySpark, Delta Lake, Databricks, ThreadPool, ROUGE/BLEU evaluation

- Processed 5M customer feedback entries across 5 domains with parallel LLM inference, **adaptive DSPy prompting**, idempotent Delta Merge, and interactive dashboards; applied LLM-as-Judge metrics for output quality evaluation

Technical Skills

- **Programming Languages:** Python (FastAPI, Pydantic), SQL
- **Generative and Agentic AI:** Agentic AI, Multi-Agent Systems, Tool/ Function Calling, Multi-Step Orchestration, MCP, RAG, Agentic RAG, LLM-as-Judge, Prompt & Context Engineering, Guardrails, Hallucination Detection, Fine-tuning, Azure OpenAI, Gemini, multimodal LLMs, Claude, OpenAI API, AWS Bedrock
- **Databricks:** Genie, Vector Search, Unity Catalog, MLflow, Delta Lake, Scale-to-Zero Model Serving
- **Agent Frameworks:** LangChain, LangGraph, LangSmith, DSPy, Pydantic AI, CrewAI, AutoGen, Hugging Face, Microsoft Foundry
- **RAG, Search and Document AI:** OCR, Layout/Table Extraction, Multimodal LLMs, Azure Document Intelligence, Pinecone, Semantic/Layout-Aware Chunking, Hybrid Search, Reranking, Query Rewriting, RAGAS
- **Evaluation and Observability:** RAGAS, ROUGE, BLEU, LangSmith, MLflow, Agent Evaluation Frameworks
- **Traditional ML and NLP:** PyTorch, Scikit-learn, XGBoost, BERT, Time Series Forecasting (ARIMA, SARIMA), Anomaly Detection, Sentiment Analysis
- **Data Engineering:** PySpark, Delta Lake, Delta Merge, Change Data Feed, Unity Catalog, ETL/ELT Pipelines, Azure Data Factory, Batch Processing
- **Cloud Platforms:** Azure (Microsoft Foundry, Document Intelligence, S3,Bedrock), AWS (Bedrock,SageMaker), Snowflake, Databricks, GCP (Vertex AI)
- **MLOps and DevOps:** Docker, Kubernetes, Terraform, Azure DevOps, MLflow, CI/CD, GitHub Actions, REST APIs, Git
- **Security:** JWT, AES-256 Encryption, PII Redaction
- **Business Analysis & Stakeholder Engagement:** Requirements Gathering, Workshop Facilitation, Technical Documentation, Status Reporting, Client-Facing Communication
- **Soft Skills:** Rapid Prototyping, Technical Leadership, Mentoring

Certifications

- **AWS Certified AI Practitioner** – Amazon Web Services, July 2026
- **Generative AI Leader Certification** – Google Cloud, July 2026
- **Professional Machine Learning Engineer** – Google Cloud, May 2026
- **Claude Code in Action** – Anthropic, Credential ID gz4ob4ade2w4, February 2026
- **AI Fluency Framework and Foundations** – Anthropic, Credential ID p2ebyk95ny2r, September 2025
- **LangGraph Certification** – LangChain Academy, Credential ID akwt3x962i, August 2025
- **Snowflake Squad 2025** – Snowflake, Credential ID 129290404, January 2025
- **AI Expert Badge** – Deloitte

Education

CVR College of Engineering

Bachelor of Technology in Electronics and Communication Engineering
CGPA: 8.24/10

Hyderabad, Telangana
2019 – 2023